Eric Rowland

Department of Mathematics
Hofstra University
Hempstead, New York 11549-1000

https://ericrowland.github.io
https://www.youtube.com/@EricRowland

Research Number theory, combinatorics, theoretical computer science.

Academic Positions

Hofstra University
Associate Professor, Department of Mathematics,
Fall 2021 – present.

Hofstra University
Assistant Professor, Department of Mathematics,
Fall 2016 – Spring 2021.

University of Liège
Marie Curie Actions COFUND Postdoctoral Fellow, Department of Mathematics,
January 2014 – December 2015.
Supervisor: Michel Rigo

University of Québec at Montréal
Postdoctoral Fellow, LaCIM, January 2012 – December 2013.
Supervisors: Srecko Brlek and Christophe Reutenauer

University of Waterloo
Postdoctoral Fellow, School of Computer Science, Fall 2011.
Supervisor: Jeffrey Shallit

Tulane University
NSF VIGRE Postdoctoral Researcher, Department of Mathematics, Fall 2009 – Spring 2011.
Supervisor: Victor Moll

Education

Rutgers University, New Brunswick
Ph.D. in Mathematics, 2009.
Advisor: Doron Zeilberger

University of California, Santa Cruz
B.A. in Mathematics with Highest Honors, 2003.

Publications

Complexity of powers of a constant-recursive sequence, with Jesús Sistos Barrón*,
submitted (23 pages).

The entries of the Sinkhorn limit of an $m \times n$ matrix, with Jason Wu**,
submitted (25 pages).

Algebraic power series and their automatic complexity modulo prime powers, with Reem Yassawi,
submitted (43 pages).

Limiting density of the Fibonacci sequence modulo powers of a prime, with Nicholas Bragman*,
Research in Number Theory **11** (2025) 88 (21 pages). Single-blinded.

An elementary proof of Bridy's theorem, with Manon Stipulanti and Reem Yassawi,
Finite Fields and Their Applications **105** (2025) 102621 (37 pages). Single-blinded.

*Single asterisks indicate undergraduate students. **Two asterisks indicate high school students.

How to referee a math paper,
Notices of the American Mathematical Society **71** (2024) 1183–1185.

A natural bijection for contiguous pattern avoidance in words, with Julia Carrigan* and Isaiah Hollars*,
Discrete Mathematics **347** (2024) 113793 (8 pages). Single-blinded.

The lexicographically least square-free word with a given prefix, with Siddharth Berera*, Andrés Gómez-Colunga*, Joey Lakerdas-Gayle*, John López, Mauditra Matin*, Daniel Roebuck*, Noam Scully*, and Juliet Whidden*,
The Electronic Journal of Combinatorics **30** (2023) #P3.11 (43 pages). Single-blinded.

Magic numbers in periodic sequences, with Savinien Kreczman, Luca Prigioniero, and Manon Stipulanti,
14th International Conference on Combinatorics on Words (WORDS 2023), *Lecture Notes in Computer Science* **13899** (2023) 206–219. Single-blinded.

Lucas' theorem modulo p^2,
The American Mathematical Monthly **129** (2022) 846–855. Double-blinded.

Ultimate periodicity problem for linear numeration systems, with Émilie Charlier, Adeline Massuir, and Michel Rigo,
International Journal of Algebra and Computation **32** (2022) 561–596. Single-blinded.

Lucas congruences for the Apéry numbers modulo p^2, with Reem Yassawi and Christian Krattenthaler,
Integers **21** (2021) A20 (15 pages). Single-blinded.

Automaticity and invariant measures of linear cellular automata, with Reem Yassawi,
Canadian Journal of Mathematics **72** (2020) 1691–1726. Single-blinded.

Avoiding 5/4-powers on the alphabet of nonnegative integers, with Manon Stipulanti,
The Electronic Journal of Combinatorics **27** (2020) #P3.42 (39 pages). Single-blinded.

Avoiding 5/4-powers on the alphabet of nonnegative integers (extended abstract), with Manon Stipulanti,
24th International Conference on Developments in Language Theory (DLT 2020), *Lecture Notes in Computer Science* **12086** (2020) 280–293. Single-blinded.

A closed-form solution might be given by a tree: valuations of quadratic polynomials, with Leyda Almodóvar, Alyssa Byrnes*, Julie Fink*, Xiao Guan*, Aashita Kesarwani, Gary Lavigne*, Luis Medina, Victor Moll, Isabelle Nogues*, Senthil Rajasekaran*, and Amber Yuan*,
SCIENTIA, Series A: Mathematical Sciences **29** (2019) 11–28. Invited; no peer review.

IntegerSequences: a package for computing with k-regular sequences,
6th International Congress on Mathematical Software (ICMS 2018), *Lecture Notes in Computer Science* **10931** (2018) 414–421. Single-blinded.

Avoiding fractional powers over the natural numbers, with Lara Pudwell,
The Electronic Journal of Combinatorics **25** (2018) #P2.27 (46 pages). Single-blinded.

A matrix generalization of a theorem of Fine,
Integers **18A** (2018) A18 (11 pages). Single-blinded.

Binomial coefficients, valuations, and words,
21st International Conference on Developments in Language Theory (DLT 2017), *Lecture Notes in Computer Science* **10396** (2017) 68–74. Invited; no peer review.

Periodicity in the p-adic valuation of a polynomial, with Luis Medina and Victor Moll,
Journal of Number Theory **180** (2017) 139–153. Single-blinded.

p-adic asymptotic properties of constant-recursive sequences, with Reem Yassawi,
Indagationes Mathematicae **28** (2017) 205–220. Single-blinded.

Profinite automata, with Reem Yassawi,
Advances in Applied Mathematics **85** (2017) 60–83. Single-blinded.

Decision algorithms for Fibonacci-automatic words, II: related sequences and avoidability, with Chen Fei Du, Hamoon Mousavi, Luke Schaeffer, and Jeffrey Shallit,
Theoretical Computer Science **657** (2017) 146–162. Single-blinded.

Infinite products arising in paperfolding, with Leyda Almodóvar, Victor Moll, Hadrian Quan*, Fernando Roman*, and Michole Washington*,
Journal of Integer Sequences **19** (2016) 16.5.1 (18 pages). Single-blinded.

What's in YOUR wallet?, with Lara Pudwell,
The Mathematical Intelligencer **37** (2015) 54–60. Single-blinded.

p-regularity of the p-adic valuation of the Fibonacci sequence, with Luis Medina,
The Fibonacci Quarterly **53** (2015) 265–271. Single-blinded.

Automatic sets of rational numbers, with Jeffrey Shallit,
International Journal of Foundations of Computer Science **26** (2015) 343–365. Single-blinded.

Automatic congruences for diagonals of rational functions, with Reem Yassawi,
Journal de Théorie des Nombres de Bordeaux **27** (2015) 245–288. Single-blinded.

Classification of automorphic conjugacy classes in the free group on two generators, with Bobbe Cooper,
Algorithmic Problems of Group Theory, Their Complexity, and Applications to Cryptography, edited by D. Kahrobaei and V. Shpilrain, *Contemporary Mathematics* **633** (2015) 13–40. Single-blinded.

What is an automatic sequence?,
Notices of the American Mathematical Society **62** (2015) 274–276. Single-blinded.

A new approach to the 2-regularity of the ℓ-abelian complexity of 2-automatic sequences, with Aline Parreau, Michel Rigo, and Élise Vandomme,
The Electronic Journal of Combinatorics **22** (2015) #P1.27 (44 pages). Single-blinded.

A characterization of p-automatic sequences as columns of linear cellular automata, with Reem Yassawi,
Advances in Applied Mathematics **63** (2015) 68–89. Single-blinded.

Structure and enumeration of $(3+1)$-free posets, with Mathieu Guay-Paquet and Alejandro Morales,
Annals of Combinatorics **18** (2014) 645–674. Single-blinded.

A case study in meta-automation: automatic generation of congruence automata for combinatorial sequences, with Doron Zeilberger,
Journal of Difference Equations and Applications **20** (2014) 973–988. Single-blinded.

The rational–transcendental dichotomy of Mahler functions, with Jason Bell and Michael Coons,
Journal of Integer Sequences **16** (2013) 13.2.10 (11 pages). Single-blinded.

Structure and enumeration of $(3+1)$-free posets (extended abstract), with Mathieu Guay-Paquet and Alejandro Morales,
25th International Conference on Formal Power Series and Algebraic Combinatorics (FPSAC 2013), *Discrete Mathematics & Theoretical Computer Science Proceedings* (2013) 253–264. Single-blinded.

Growing words in the free group on two generators, with Bobbe Cooper,
Illinois Journal of Mathematics **55** (2011) 417–426. Single-blinded.

Boundary growth in one-dimensional cellular automata, with Charles Brummitt,
Complex Systems **21** (2012) 85–116. Single-blinded.

The iterated integrals of $\ln(1 + x^n)$, with Tewodros Amdeberhan, Christoph Koutschan, and Victor Moll,
International Journal of Number Theory **8** (2012) 71–94. Single-blinded.

Avoiding 3/2-powers over the natural numbers, with Jeffrey Shallit,
Discrete Mathematics **312** (2012) 1282–1288. Single-blinded.

Toward a language theoretic proof of the four color theorem, with Bobbe Cooper and Doron Zeilberger,
Advances in Applied Mathematics **48** (2012) 414–431. Single-blinded.

k-automatic sets of rational numbers, with Jeffrey Shallit,
6th International Conference on Language and Automata Theory and Applications 2012, *Lecture Notes in Computer Science* **7183** (2012) 490–501. Single-blinded.

The number of nonzero binomial coefficients modulo p^α,
Journal of Combinatorics and Number Theory **3** (2011) 15–25. Single-blinded.

Iterated primitives of logarithmic powers, with Luis Medina and Victor Moll,
International Journal of Number Theory **7** (2011) 623–634. Single-blinded.

Pattern avoidance in binary trees,
Journal of Combinatorial Theory, Series A **117** (2010) 741–758. Single-blinded.

Non-regularity of $\lfloor \alpha + \log_k n \rfloor$,
Integers **10** (2010) 19–23. Single-blinded.

Counting interesting elections, with Lara Pudwell,
The American Mathematical Monthly **117** (2010) 167–174. Single-blinded.

Regularity versus complexity in the binary representation of 3^n,
Complex Systems **18** (2009) 367–377. Single-blinded.

A natural prime-generating recurrence,
Journal of Integer Sequences **11** (2008) 08.2.8 (13 pages). Single-blinded.

Local nested structure in rule 30,
Complex Systems **16** (2006) 239–258. Single-blinded.

YouTube Channel

Videos about surprising results in number theory (9 million views),
`https://www.youtube.com/@EricRowland`.

Software Development

16 Mathematica packages for computation and automated theorem proving in a variety of areas,
`https://ericrowland.github.io/packages.html`.

Wolfram|Alpha
R&D Fellow / Consultant, 2006 – 2018.
Design and implementation of Wolfram|Alpha's framework for answering geometry questions.
Additional functionality for number theory, cellular automata, music theory, graphs, polyhedra, etc.

Grants and Awards

REU Site: New York City Discrete Mathematics REU, 2024–2026,
NSF award DMS-2349366. Senior Personnel; Principal Investigator: Adam Sheffer.

Finalist, Summer of Math Exposition 2
for my video "1 Billion is Tiny in an Alternate Universe: Introduction to p-adic Numbers",
`https://www.3blue1brown.com/blog/some2`, 2022.

Award for Distinguished Teaching,
Metro New York Section of the Mathematical Association of America, 2021.

REU Site: New York City Discrete Mathematics REU, 2021–2023,
NSF award DMS-2051026. Senior Personnel; Principal Investigator: Adam Sheffer.

HCLAS Faculty Research and Development Grants,
Hofstra University, five awards 2018–2025.

BeIPD-COFUND-IPD Grant,
University of Liège and the European Union, January 2014 – December 2015.

NSF U.S. Junior Oberwolfach Fellow,
Travel grant to attend Combinatorics on Words workshop, Oberwolfach, Germany, August 2010.

Graduate Assistantship, Rutgers University, Spring 2009.

Teaching Excellence Award, Rutgers University, Spring 2006.

Henry C. Torrey Graduate Fellowship, Rutgers University, Fall 2003 – Spring 2005.

Teaching Experience

Hofstra University
Mathematical Excursions; Linear Mathematics and Matrices; Logic, Sets, and Probability; Precalculus; Analytic Geometry and Calculus I; Introduction to Higher Mathematics; Elementary Differential Equations; Number Theory; Combinatorics and Graph Theory; Engineering Mathematics I; Experimental Mathematics

Tulane University
Calculus II; Linear Algebra; Experimental Mathematics; Abstract Algebra I

Rutgers University
Calculus II for the Mathematical and Physical Sciences; Introductory Linear Algebra

Lecture Series

University of Liège
Lectures on Mathematica as a research tool,
Liège, Belgium, Fall 2015.

MSRI Undergraduate Program
Lectures on automatic sequences, regular sequences, p-adic numbers, valuations, and experimental mathematics,
MSRI, Berkeley, CA, Summer 2014.

Wolfram Science Summer School
Lectures on cellular automata, finite automata, randomness and complexity, experimental mathematics, and programming in Mathematica,
Seven summers 2007–2016 at University of Vermont, Burlington, VT; Istituto di Scienza e Tecnologie, Pisa, Italy; Curry College, Milton, MA; and Bentley University, Waltham, MA.

Talks

Plenary Talks
Mid-Atlantic Algebra, Geometry, and Combinatorics Workshop, CUNY Graduate Center, New York, NY (2025–3–22).
Developments in Language Theory, University of Liège, Belgium (2017–8–11).
Spring Meeting of the Indiana Section of the Mathematical Association of America, Franklin College, Franklin, IN (2010–4–10).

Conference Talks

AMS Fall Southeastern Sectional Meeting, Tulane University, New Orleans, LA (2025–10–4).
SIAM Conference on Applied Algebraic Geometry, University of Wisconsin, Madison, WI (2025–7–7).
Numeration, Utrecht University, Netherlands (2024–6–4).
Applications of Computer Algebra, online (2021–7–24).
Joint Mathematics Meetings, Denver, CO (2020–1–15).
AMS Fall Southeastern Sectional Meeting, University of Florida, Gainesville, FL (2019–11–2).
International Congress on Mathematical Software, University of Notre Dame, South Bend, IN (2018–7–27).
SIAM Conference on Discrete Mathematics, University of Colorado, Denver, CO (2018–6–7).
Combinatorial and Additive Number Theory, CUNY Graduate Center, New York, NY (2018–5–25).
Joint Mathematics Meetings, San Diego, CA (2018–1–10).
Applications of Computer Algebra, Jerusalem College of Technology, Jerusalem, Israel (2017–7–20).
AMS Spring Eastern Sectional Meeting, Hunter College, CUNY, New York, NY (2017–5–6).
Bridges Between Automatic Sequences, Algebra, and Number Theory, CRM, Montréal, QC (2017–4–27).
Joint Mathematics Meetings, Seattle, WA (2016–1–7).
SIAM Conference on Applied Algebraic Geometry, NIMS, Daejeon, South Korea (2015–8–6).
Discrete Mathematics Day, University of Liège, Belgium (2015–1–8).
AMS Fall Eastern Sectional Meeting, Halifax, NS (2014–10–19).
Journées Montoises d'Informatique Théorique, Nancy, France (2014–9–24).
Challenges in Combinatorics on Words, Fields Institute, Toronto, ON (2013–4–26).
Joint Mathematics Meetings, San Diego, CA (2013–1–9).
ISM Quebec Student Conference, UQAM, Montréal, QC (2012–6–1).
Combinatorics, Automata, and Number Theory, Marseille, France (2012–5–23).
Language and Automata Theory and Applications, A Coruña, Spain (2012–3–5).
Outstanding Challenges in Combinatorics on Words, Banff, AB (2012–2–20).
Joint Mathematics Meetings, Boston, MA (2012–1–6).
Permutation Patterns, California Polytechnic State University, San Luis Obispo, CA (2011–6–22).
Joint Mathematics Meetings, New Orleans, LA (2011–1–8).
Mini-Workshop: Combinatorics on Words, Oberwolfach, Germany (2010–8–27).
From $A = B$ to $Z = 60$, Piscataway, NJ (2010–5–27).
Joint Mathematics Meetings, San Francisco, CA (2010–1–14).
Joint Mathematics Meetings, Washington, DC (2009–1–5).
Midwest NKS Conference, University of Indiana, Bloomington, IN (2008–10–31).
Joint Mathematics Meetings, San Diego, CA (2008–1–6).
NKS2007 Conference, University of Vermont, Burlington, VT (2007–7–14).
NKS2006 Conference, Washington, DC (2006–6–17).
Midwest NKS Conference, University of Indiana, Bloomington, IN (2005–10–28).
NKS2004 Conference, Boston, MA (2004–4–25).

60 Seminar Talks

CUNY Graduate Center, Hofstra University, INRIA Rocquencourt, The Open University, Queen's University, RISC (Austria), Rutgers University, Trent University, Tulane University, UC Davis, UCLA, UIUC, University of Liège, UMass Amherst, UQAM, University of Waterloo, Valparaiso University, William Paterson University, One World Seminar on Combinatorics on Words.
Full list: `https://ericrowland.github.io/talks.html#seminars`

Outreach Talks
Columbia Grammar & Preparatory School, New York, NY (2014–10–8).
Princeton Friends School, Princeton, NJ (2008–11–15).
Guest "Problem of the Week" lecture, Princeton Friends School, Princeton, NJ (2005–4–29).

Postdoctoral Mentee

Manon Stipulanti, Visiting Scholar,
Hofstra University, October 2019 – August 2020.

Graduate Students

Allison Siegel, Master's thesis,
Hofstra University, 2026.

Philip Kopel, Master's thesis: Classical and recent advances in fast polynomial multiplication,
Tulane University, 2011.

Undergraduate Research

August Fogler, Contiguous pattern avoidance in arrays,
New York City Discrete Mathematics REU, Baruch College, Summer 2025.

Aidan Hackett, Algebraic complexity of factorial ratios,
New York City Discrete Mathematics REU, Baruch College, Summer 2025.

Hamza Virk,
Hofstra University, Spring 2025 – Fall 2025.

Najalia Singh, Honors Thesis: Lucas congruences modulo p^2 for the Apéry numbers associated with $\zeta(2)$,
Hofstra University, Spring 2024.

Megan Paasche, The Goulden–Jackson cluster method for pattern avoidance in binary trees,
New York City Discrete Mathematics REU, Baruch College, Summer 2023.

Jesús Sistos Barrón, Complexity of powers of a constant-recursive sequence,
New York City Discrete Mathematics REU, Baruch College, Summer 2023.

Julia Carrigan and Isaiah Hollars, A natural bijection for contiguous pattern avoidance in words,
New York City Discrete Mathematics REU, Baruch College, Summer 2022.

Joshua Crisafi, Honors Thesis: Lucas congruences modulo p^2 for a variety of integer sequences,
Hofstra University, Fall 2021 – Spring 2022.

22 students worldwide, The lexicographically least square-free word with a given prefix,
Polymath Jr., Summer 2021.

Daniel Dimijian, Searching the Apéry numbers efficiently,
Hofstra University, Spring 2020 – Fall 2020.

Brandon Crofts, Honors Thesis: An exploration and generalization of the KRC sequence,
Hofstra University, Fall 2019 – Spring 2020. This thesis won the university's 2020 Undergraduate Library Research Award in the sciences category.

Mike Chinbayar, Honors Thesis: Pascal's triangle and finding the expected value,
Hofstra University, Fall 2019.

Kawkab Abid, The Collatz function in base 3,
Hofstra University, Spring 2019 – Fall 2019.

Nicholas Bragman, Limiting density of the Fibonacci sequence modulo powers of p,
Hofstra University, Fall 2017 – Summer 2019.

Brian Zilli, Honors Thesis: Some properties of reversible cellular automaton rules,
Hofstra University, Spring 2018.

6 undergraduate research teams (18 students), co-supervised with Victor Moll,
MSRI-UP, Berkeley, CA, 2014.

34 individual research projects,
Wolfram Science Summer School, seven summers 2007–2016.

High School Student Research

Maxwell Lippmann,
Spring 2025 – Fall 2025.

Arav Chand, Universal matrices for counting Fibonomial and C-nomial coefficients by their p-adic valuations,
Summer 2023 – Fall 2025. In 2024, Arav won 1st place in the Math, Physics, & Astronomy category at the Long Island Science & Engineering Fair and won the 3rd place NSA Research Directorate Award in the mathematics category at the International Science and Engineering Fair. In 2025, Arav again won 1st place at the Long Island Science & Engineering Fair in his category (now Robotics, Intelligent Machines, Software, and Math) and was among the Grand Award winners (Fourth Award) in Mathematics at the International Science and Engineering Fair.

Jason Wu, Sinkhorn limits of 3×3 matrices,
Summer 2022 – Spring 2024. Jason won 1st place in the Mathematics category at the 2023 New York State Science & Engineering Fair and went on to participate in the International Science and Engineering Fair.

Yousuf Kidwai, A substitution system for cellular automaton rule 2230,
Summer 2019.

Kevin Chen and Abel Varghese, Experimental determination of Sinkhorn limits,
Hofstra University Summer Science Research Program, Summer 2019.

Jared Bank, An analysis of boundary growth rates in one-dimensional cellular automata,
Fall 2018. Jared won 1st place in the math category at the 2019 Long Island Science & Engineering Fair and went on to participate in the International Science and Engineering Fair.

Anna Finkelstein, Reversibility in cellular automata,
Summer 2017.

Conference & Seminar Organization

New York Combinatorics Day,
Hofstra University, April 2024.

Special Session on Experimental Mathematics in Number Theory and Combinatorics,
AMS Fall Southeastern Sectional Meeting, online, November 2021.

Annual Meetings of the Metro New York Section of the Mathematical Association of America,
2020 – present.

AMS Special Session on Sequences, Words, and Automata,
Joint Mathematics Meetings, Denver, CO, January 2020.

Hofstra Mathematics Seminar,
Hofstra University, Fall 2019 – Spring 2020.

New York Combinatorics Seminar,
2018 – present.

Annual Meeting of the Metro New York Section of the Mathematical Association of America,
Hofstra University, May 2018.

AMS Special Session on Arithmetic Properties of Sequences from Number Theory and Combinatorics,
Joint Mathematics Meetings, Atlanta, GA, January 2017.

Workshop on Automatic Sequences,
University of Liège, May 2015.
This conference was funded by my BeIPD-COFUND-IPD grant and had 50 participants.

Languages and Automata Theory Seminar,
University of Waterloo, Fall 2011.

From $A = B$ to $Z = 60$, a conference celebrating Doron Zeilberger's 60th birthday,
DIMACS, Rutgers University, May 2010.

Round Table Discussions Seminar (co-founding organizer),
Rutgers University, Fall 2008 – Spring 2009.

Professional Service

Committee on Section Bylaws, Mathematical Association of America
2025 – present.

Secretary, Metropolitan New York Section of the Mathematical Association of America,
2024 – present.

Thesis Committee for Adeline Massuir's Ph.D. Defense,
University of Liège, April 2021.

Prize Committee, Wolfram Rule 30 Prizes (totaling $30,000),
2019 – present.

Webmaster, Metropolitan New York Section of the Mathematical Association of America,
2019 – present.

Editorial Board, *Complex Systems*,
2018 – present.

Wikipedia editor (700 edits, modifying 400 pages) and initial creator of the articles Alan Cobham (mathematician), Constant-recursive sequence, k-regular sequence, and Legendre's formula.

Programmatic checking and correcting of the OEIS database,
Summer 2017 and Summer 2022.

Reviewer, 99 articles.
Journals: *Acta Arithmetica*, *Advances in Applied Mathematics*, *The American Mathematical Monthly*, *Annals of Combinatorics*, *Bulletin of the Australian Mathematical Society*, *Combinatorica*, *Complex Systems*, *Discrete Mathematics*, *Discrete Mathematics Letters*, *The Electronic Journal of Combinatorics*, *European Journal of Combinatorics*, *Functiones et Approximatio*, *Graphs and Combinatorics*, *Indagationes Mathematicae*, *Indian Journal of Pure and Applied Mathematics*, *Integers*, *Journal de Théorie des Nombres de Bordeaux*, *Journal of Algebraic Combinatorics*, *Journal of Combinatorial Theory Series A*, *Journal of Combinatorics and Number Theory*, *Journal of Difference Equations and Applications*, *Journal of Integer Sequences*, *Journal of Number Theory*, *Journal of Symbolic Computation*, *The Mathematica Journal*, *Mathematics Exchange*, *Mathematics Magazine*, *Proceedings of the American Mathematical Society*, *Publicationes Mathematicae Debrecen*, *RAIRO Theoretical Informatics and Applications*, *Research in Number Theory*, *Séminaire Lotharingien de Combinatoire*, *Theoretical Computer Science*.
Conferences: Developments in Language Theory (DLT), International Colloquium on Automata, Languages, and Programming (ICALP), International Congress on Mathematical Software (ICMS), International Symposium on Algorithms and Computation (ISAAC), International Symposium on Symbolic and Algebraic Computation (ISSAC), International Workshop on Combinatorial Algorithms (IWOCA), Language and Automata Theory and Applications (LATA), WORDS.

Author, 32 reviews of articles for *Mathematical Reviews*.

Reviewer, 2 grants submitted to government funding agencies.

External reviewer, Best Paper Award of TU Vienna's Faculty of Mathematics.

University Service

HOFSTRA UNIVERSITY

Admission events (Admitted Student Day / Fall Open House / First-Year Student Fair), February 2017, October 2017, October 2018, October 2019 (two), April 2020, November 2020, January 2021, February 2022, April 2023.

Hiring Committee for Web and Mobile Developer position, Spring 2020.
Honor Board, Fall 2019 – Spring 2025.
Academic Calendar Committee, Spring 2018 – Spring 2022.
Mathematics Club Content Advisor, Fall 2017 – present.

Departmental Service

HOFSTRA UNIVERSITY

Departmental Personnel Committee, Fall 2023.
Departmental Personnel Committee, Fall 2021.
Mathematics Department Tech Liaison, Fall 2017 – present.
Mathematics Department Library Liaison, Spring 2017 – present.
Internship Committee, Spring 2017.
Hiring Committee, Spring 2017.

RUTGERS UNIVERSITY

Organizer, Directed Reading Program for undergraduates, Fall 2005 – Spring 2009.

Updated November 6, 2025